Product Requirements Document (PRD)

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Component 0: Drone Types & Manufacturing Scope

0.1 3 Major Drone Types to be Manufactured

Drone Type	Key Features	Primary Applications in Mass Manufacturing
Multi-Rotor	Multiple propellers (quad-, hexa-, octocopters) provide excellent stability and manoeuvrability, and they can hover with precision (small to mid-level distances with multi-directional journeys & low payload carrying capacity).	* Inspection: Power lines, pipelines, and industrial infrastructure. * Photography/Videography: Real estate, media, and mapping. * Surveillance & Security (Internal): Facility monitoring and perimeter security (CISF & CRPF)
Fixed-Wing	Resemble airplanes; use forward momentum for lift, making them energy-efficient for long distances (with medium payload carrying capacity).	* Surveying & Mapping: Large-scale land surveys and GIS applications (thousands of acres). * Agriculture & Animal Husbandry: Monitoring large farmlands and crop health analysis. Selective Fertilizer & Vaccine Delivery. * Disaster Response: Wildfire tracking and oil spill detection due to long flight times. Mini water purification system (SWaDeS) delivery in earthquake hit zones.
Single-Rotor	Similar to traditional helicopters with one large main rotor and a tail rotor. Highly efficient with longer endurance, especially when gas-powered. Can also be a hybrid, solar battery (chargeable) + gas powered	* Light-Medium Payload Transport: Capable of carrying small tooling & sensors or other specialized micro industrial equipment. * Long-Duration Missions: Used in construction, zonal Défense, chemical, petrochemical, and energy sectors where endurance is critical.

0.2 Major Industrial Procedures & Scope

A. Frame & Structural Component Fabrication

2 teams of 5 employees. 2 per machine. 1 relief man.

- **Injection Moulding Machines ($\times$ 2) (*partial in-house*)**: High-volume production of plastic frames, housings, and propellers for special-purpose lightweight, consistent parts.
- **CNC Machining Centers (Mills, Lathes, 5-Axis Machines) ($\times$ 5) (*in-house*)**: Precision cutting of metal or composite parts (e.g., aluminium mounts, carbon fiber arms). 2 machines active at any given point. 1 on alternative reserve (activated after every 12 days). The other 2 machines are operated as emergency reserve (if no emergency, brought into manufacturing after every 16 hours).
- **Laser Cutting Machines or Water Jets (*partial in-house*)**: Shaping flat materials like carbon fiber sheets or metals.
  (Manual layering & curing of carbon fiber sheets in autoclaves & vacuum bagging are outsourced)

B. Main Assembly Line + Electronics Motor, & Propulsion Assembly, Fixtures, & Integration (PCB's & Components)
Semi-automated with skilled employees. 4 teams of 6. 2 teams working parallelly per shift. 5 major operators. 1 reserve (relief man). 2 + 1-line engineers.

- **Surface Mount Technology (SMT) Lines:** Automated pick-and-place machines, soldering stations, flight controller assembly, ESCs (electronic speed controllers), sensors (accelerometers & selective gyroscopes), & cameras.
- **Motor Winding & Assembly Stations:** Attaching motors (servo & induction), magnets, bearings, & test propulsion systems.
- **Robotic Arms & Conveyor Systems:** Integrating frame, motors, propellers, batteries, cameras/gimbals, and electronics. Final soldering stations, screwdrivers, and adhesive applicators, wiring, flight-controller mounting for secure connections. Sensor Integration in clean-room environments (supervised). MQTT integration (*self-cloud*).

C. Testing and Quality Control, Packaging, & Shipping:

- **Environmental Test Chambers (*outsourced*):** Simulating temperature, humidity, and vibration.
- **Non-Destructive Testing (NDT) Equipment (*in-house*):** Ultrasonic devices, thermal cameras, and X-ray for defect detection.
- **Flight Test Rigs and Wind Tunnels (*outsourced*):** Calibration of motors, stability checks, sensor verification, and controlled flight trials.
- **Avionics Testing Tools (*partial embedded application*):** *Software flashing, firmware uploads, and functional checks*.
- **Packaging and Shipping (*partial in house*)**: Automated packaging lines for final boxing, labelling, and quality sampling.

Component 1: Application of Frameworks

1.1. Jobs-To-Be-Done (JTBD) Framework

Primary Job

When running daily drone electronics manufacturing operations, Operations and Production leaders want to anticipate issues and dynamically adapt production, so they can meet delivery timelines, control costs, maintain quality, and retain top-skilled staff without firefighting.

Supporting Jobs

- Detect machine failures before breakdowns disrupt schedules
- Reprioritize production quickly when urgent or high-value orders arrive
- Identify quality defects early to avoid downstream rework
- Ensure critical components are available just in time
- Balance workforce load without overdependence on overtime
- Selective prep-automation between sessions to reduce assembly & fabrication login starts

Emotional & Social Jobs

- Feel confident and in control during high-pressure production scenarios
- Reduce stress caused by last-minute escalations and manual coordination
- Build credibility with leadership through predictable delivery and measurable improvements
- Execute multiple orders in time & 2-3 varieties/variants simultaneously if needed in due time

1.2. User Persona and Pain Points

- **Operations Manager (Persona):** A mid-30s manufacturing professional in an industrial hub. Responsible for overseeing production schedules and machine uptime.

  Needs: Real-time visibility into production; automated alerts for equipment issues; ability to adapt schedules quickly.

  Pain Points: Late detection of bottlenecks, manual crisis management, missed delivery deadlines.

  Goals: Minimize machine & operator downtime, meet on-time delivery commitments, and ensure required compliance.

- **Supply Chain Manager (Persona):** A 30-42 year-old professional managing procurement in a drone plant.

  Needs: Accurate demand forecasting, supplier performance insights, and automated reorder triggers.

  Pain Points: Vendor delays, stockouts, lack of inventory visibility, unstable reserve stocking of critical components.

  Goals: Maintain just-in-time inventory (external), avoid shortages (internal), and reduce overall costs.

- **Production Line Supervisor (Persona):** A late-20s to early-40s supervisor on the assembly line.

Needs: Efficient shift scheduling, real-time quality dashboards with updates & colour coding, and clear task assignments for workers.

Pain Points: Manual scheduling errors causing idle time, high defect/rework rates, stress of meeting deadlines.

Goals: Balance workforce load, reduce defects, and improve throughput.

User Stories

- **Operations Manager:** *"As an Operations Manager, I want AI to predict machine failures and auto-reschedule production so that I can minimize downtime and deliver drones on time."*
- ***Acceptance Criteria**:* The AI dashboard predicts failures ≥48 hours in advance; production jobs auto-reschedule within 5 minutes of an alert; delivery timelines update in real time for clients.
- For urgent orders, the AI dashboard predicts failures ≥6 hours in advance and automates workflow to ensure production completion at least until the current order gets executed.

- **Supply Chain Manager:** *"As a Supply Chain Manager, I want AI to forecast component demand and vendor risks so that I can ensure just-in-time availability of drone parts."*
- ***Acceptance Criteria**:* Demand forecasts update weekly with ≥85% accuracy; vendor risk alerts trigger ≥3-7 days before a predicted delay (3 for specific components, 7 for generic bulk components); procurement requests auto-generate when stock falls below safety thresholds.
- For orders coming in at a war footing, vendor risk alerts trigger ≥36-48 hours before a predicted delay in critical components.

- **Production Line Supervisor:** *"As a Line Supervisor, I want AI to run real-time quality checks at both servicing station speeds & product quality after each stage and optimize task allocation so that I can reduce defects and balance worker load."*
- ***Acceptance Criteria**:* Computer vision flags ≥95% of PCB/soldering defects on the line; tasks auto-allocate based on worker skills and shift schedules; shift dashboard refreshes every 60 seconds with throughput and defect metrics.

1.3. Problem Frequency and Severity

Problem	Frequency / Occurrence	Severity	Detection	Risk Priority
Unplanned machine downtime	Recurring (weekly/monthly)	High	Medium-High	Medium
Manual rescheduling during order changes	Daily	High	High	Medium-High
Late quality defect detection	Daily	Medium–High	Medium-High	High
Inventory shortages or vendor delays	Monthly	High	High	High
Workforce imbalance	Daily	Medium	Medium-High	Medium-High

These issues are systemic, recurring, and escalate rapidly under urgent or custom drone orders, directly impacting delivery timelines and cost.

1.4. Key Success Metrics and Measurement

Operational Efficiency

- Unplanned downtime reduction: ≥30%
  Measured via machine and maintenance logs within 90 days
- AI-driven schedule adjustments: ≥40% of batches auto-rescheduled
  Measured via scheduling engine logs within 60 days

Quality

- Defect and rework reduction: ≥25%
  Measured via inspection and rejection data within 90 days

Supply Chain Reliability

- On-time availability of critical components: ≥20% improvement
  Measured via ERP inventory and procurement data within 120 days

Cost and Sustainability

- Energy consumption per batch: ≥15% reduction
  Measured via machine-level energy monitoring within 6 months

Adoption

- Weekly active usage by target users: ≥70%
  Measured via system usage analytics

Component 2: Value Proposition

2.1. Value Proposition Canva

Customer Profile

Customer Jobs

- Run daily drone electronics manufacturing without disruptions
- Meet delivery timelines, especially for urgent or priority orders
- Maintain quality while controlling cost and energy usage
- Coordinate machines, people, and inventory with limited real-time visibility
- In case of irregular orders, co-ordinate machine time, usage, & functionality for minimal service & maintenance.

Pains

- Unplanned machine downtime causing cascading delays
- Manual rescheduling during order changes and breakdowns
- Late detection of quality defects leading to rework & mid-stream bottlenecks
- Stockouts of critical components and delayed vendor response
- High stress and firefighting during priority orders

Gains

- Predictable production outcomes
- Early warning before failures or shortages
- Ability to reprioritize production confidently
- Fewer escalations and less manual coordination
- Credibility with leadership through measurable performance improvements
- Goal driven workflows for high-efficiency staff retention

Value Map

Products & Services

- Agentic AI platform for drone electronics manufacturing operations
- Predictive maintenance, dynamic scheduling, AI vision quality checks
- Supply chain forecasting and automated procurement triggers
- Eventually a 3-speed production system, once multiple orders start coming in for the same manufacturer/assembler (post 15-18 months, paid upgrade)

Pain Relievers

- Predicts machine failures before breakdowns occur (at semi-automated stations)
- Auto-reschedules production in response to real-time conditions
- Flags defects early on the line to avoid downstream rework
- Anticipates component shortages and vendor risks

Gain Creators

- Higher uptime and throughput
- Faster response to urgent or high-value orders

- Lower rework, energy use, and operational cost
- One operational view across machines, inventory, and workforce (uniformity & standards)

What exact problem does your product solve?

- It eliminates reactive firefighting in drone manufacturing by predicting disruptions and dynamically adjusting production, quality, and supply chain decisions in real time.

Who is the primary user of your product?

Operations Managers in drone electronics manufacturing plants. Secondary users include Production Supervisors and Supply Chain Managers.

What is the core value your product delivers?

Predictable, resilient manufacturing operations with fewer disruptions, lower cost, and higher on-time delivery. *Can handle high loads with sufficient variability.*

Why does this problem matter to the user?

Because unplanned downtime, late defects, and stockouts directly threaten delivery commitments, revenue, and professional credibility.

How is your solution better than existing alternatives?

Unlike dashboards or rule-based systems, DroneOps AI predicts issues and takes action automatically, while keeping humans in control.

What is your one-line value proposition statement?

DroneOps AI helps manufacturing teams prevent disruptions and deliver drones on time through predictive, self-adjusting operations.

Why would a user choose your product today?

It delivers measurable improvements in downtime, quality, and delivery within weeks, without replacing existing ERP or MES systems.

What user pain does your product reduce the most?

Unplanned downtime and last-minute production chaos caused by late visibility and manual coordination.

What outcome does the user achieve using your product?

Higher uptime, fewer defects, reliable deliveries, and reduced operational stress, even during urgent or priority orders.

What assumption are you making about user value?

That operations leaders value predictability and control more than raw automation or reporting dashboards.

How did you validate that this value matters?

Through recurring pain points identified in the PRD: frequent downtime, daily rescheduling, quality rework, and supply delays with clear cost impact.

What happens if this value proposition fails?

The product becomes another monitoring tool, adoption drops, and users revert to manual firefighting without trusting AI-driven actions.

Which feature directly supports your value proposition?

Predictive maintenance combined with AI-driven dynamic production scheduling.

What trade-offs did you make to focus on this value?

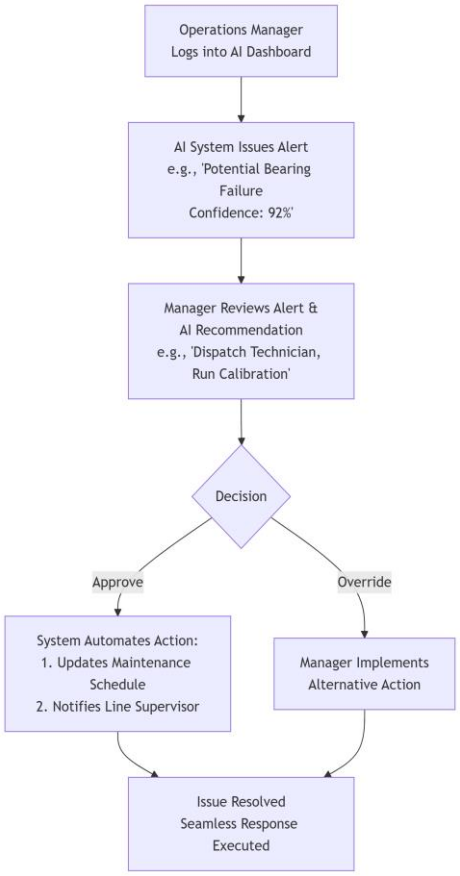
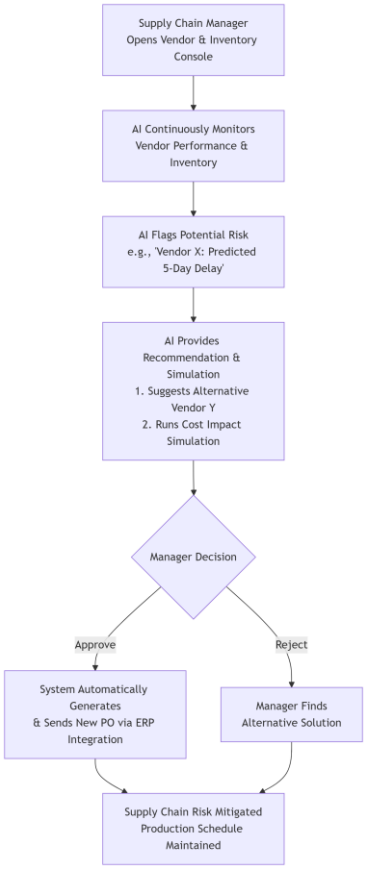
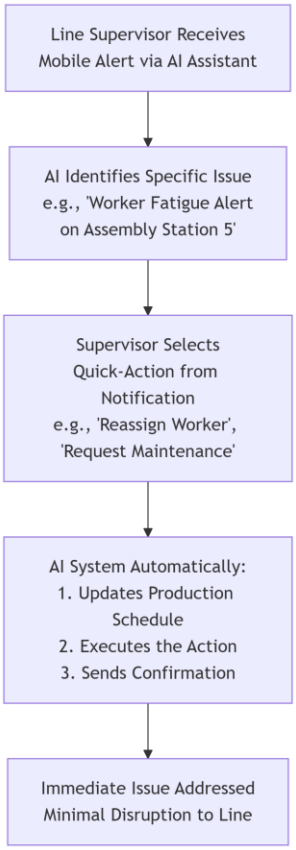
Deferred advanced analytics, digital twins, and non-core BI features to prioritize disruption prevention and operational control.

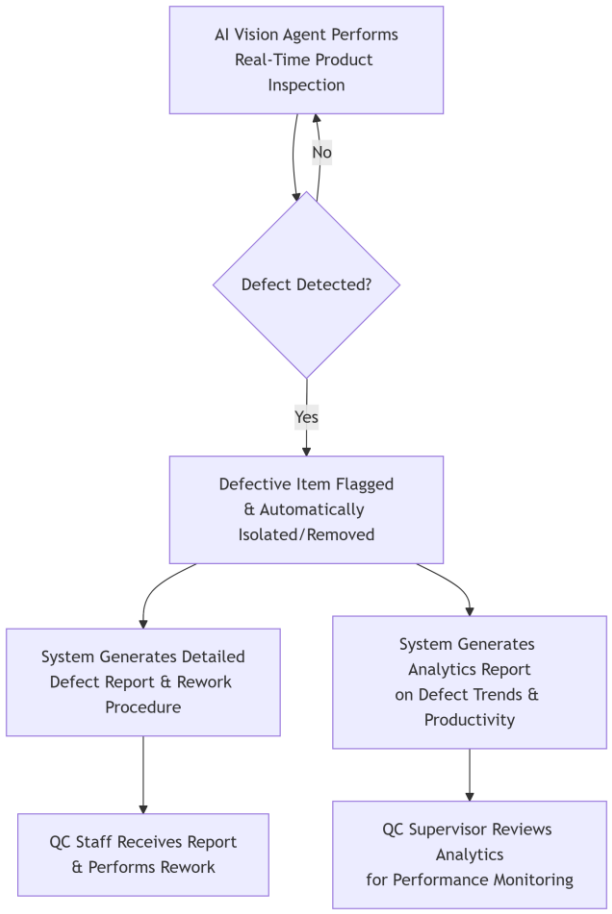
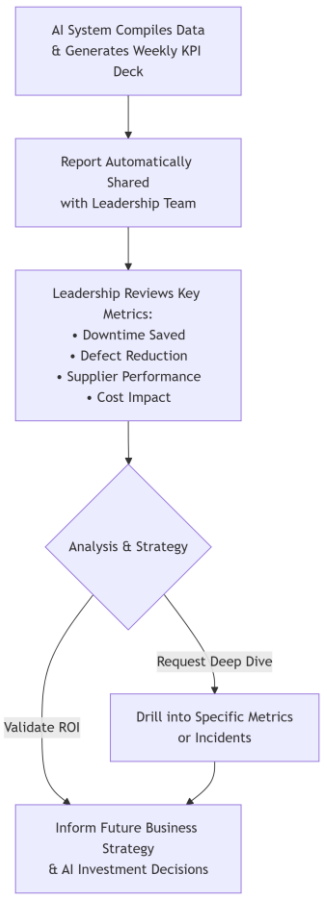
How will you measure whether users actually get this value?

- Reduction in unplanned downtime
- Percentage of AI-driven schedule adjustments
- On-time delivery rate improvement
- Reduction in defects and rework
- Weekly active usage by operations leaders

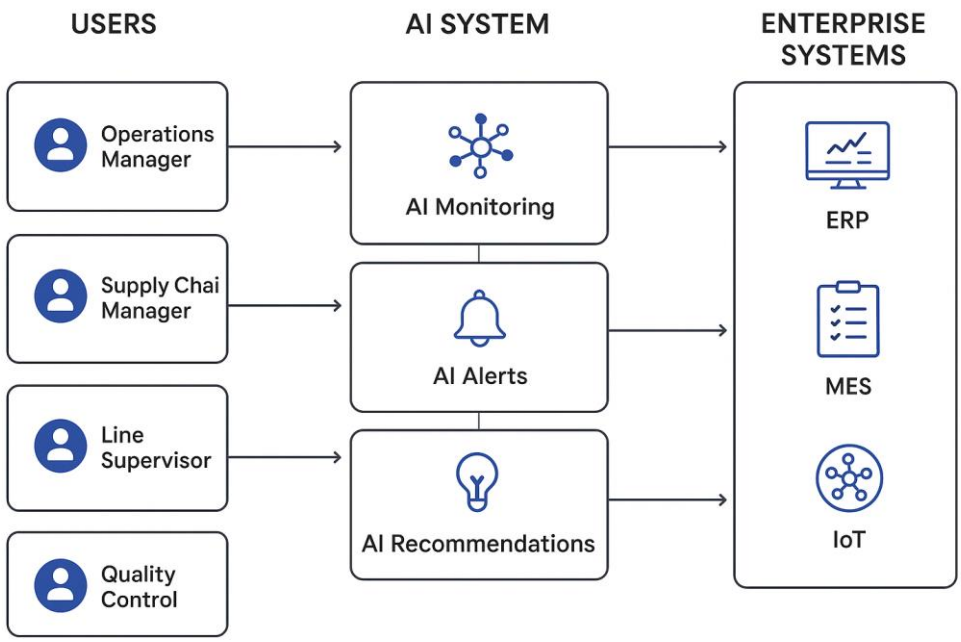
Component 3: MVP Design

3.1. User Flow Design

Operations Manager Workflow	Supply Chain Manager Workflow	Line Supervisor Workflow
		

Quality Control Workflow	Leadership Workflow
	

3.2. Information Architecture



3.3. Wireframes

Note: For wireframes created through FIGMA MAKE please refer section Component 7: Demo/Prototype

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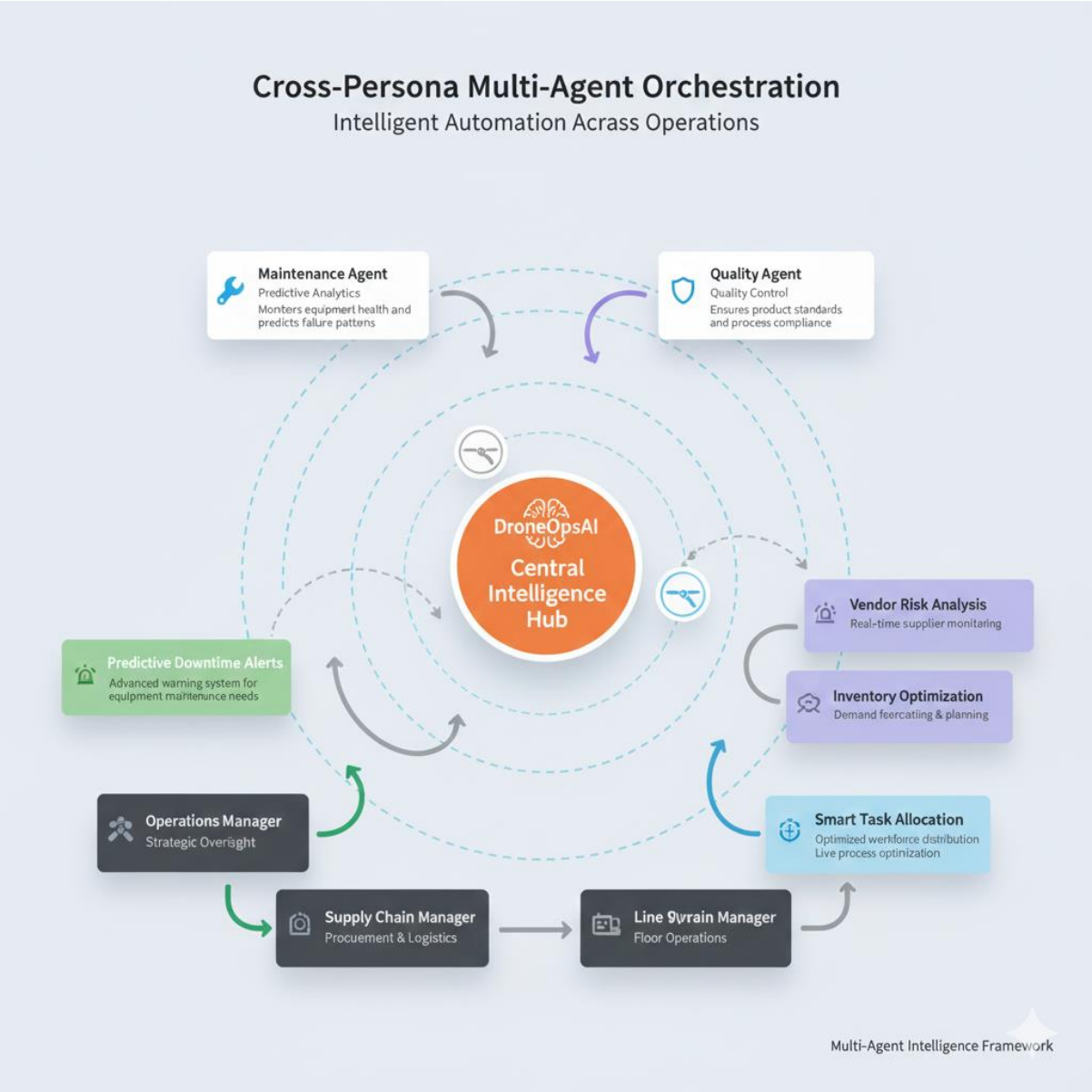


Component 4: Project Plan

4.1. Project Objective Document

Objective
Design, build, and deploy DroneOps AI as an agentic operations platform for drone electronics manufacturing that reduces unplanned downtime, improves delivery reliability, and increases operational predictability through AI-driven decision support and automation.

Business Outcome

- Fewer production disruptions
- Faster response to urgent orders
- Measurable cost and quality improvements

Target Users
Operations Managers, Production Line Supervisors, and Supply Chain Managers.

4.2. Problem Scope Definition

Problems Addressed

- Reactive handling of machine failures and production bottlenecks
- Manual and error-prone production rescheduling
- Late detection of quality defects (stage-wise production)
- Inventory shortages and delayed vendor risk visibility

Problems Not Addressed

- Drone hardware design or R&D
- Non-electronics manufacturing processes
- Enterprise ERP replacement

4.3. In-Scope Feature List

- Real-time production and machine visibility dashboards
- Predictive maintenance for critical equipment
- AI-driven dynamic production scheduling and auto-rescheduling
- Computer vision–based quality inspection
- Supply chain demand forecasting and vendor risk alerts
- Automated procurement triggers for critical components
- Human-in-the-loop controls and audit logs

4.4. Out-of-Scope Feature List

- Drone hardware or mechanical manufacturing automation
- Any features explicitly marked “Could Have” (e.g., full digital twin, advanced BI dashboards) in the initial MVP.
- Custom ERP or MES development
- Robotics control and autonomous machinery operation
- Marketing, sales, or finance analytics

4.5. Success Criteria Definition

Operational

- ≥30% reduction in unplanned downtime
- ≥25% reduction in defects and rework
- ≥20% improvement in on-time component availability

Adoption

- ≥70% weekly active usage by target users
- Successful pilot-to-rollout conversion

Business

- Clear ROI demonstrated within 90 days of pilot deployment

4.6. Project Timeline

Total Duration: 18 months

- **Phase 1 (0–6 months):** MVP development and pilot deployment

Feature / User Story	Priority	Effort (S/M/L)	Dependencies	MoSCoW
Real-time machine & production visibility (dashboards)	High	L	IoT sensors, ERP/MES integration	Must Have
Predictive maintenance for critical machines (pick-and-place, reflow oven)	High	L	Sensor data + ML models	Must Have
Dynamic production scheduling & auto-rescheduling	High	L	Scheduling engine, OM-1	Must Have
Computer vision quality checks for PCB soldering & controllers	High	L	Camera hardware + defect dataset	Must Have

- **Phase 2 (6–12 months):** Supply chain and workforce expansion

Feature / User Story	Priority	Effort (S/M/L)	Dependencies	MoSCoW
Demand forecasting for drone components (MCUs, sensors, batteries)	High	M	Historical sales + vendor data	Must Have
Auto-trigger procurement for just-in-time inventory	Medium	M	SCM-1, procurement system API	Should Have
Vendor risk scoring & alerts	Medium	M	Vendor performance data feeds	Should Have

Feature / User Story	Priority	Effort (S/M/L)	Dependencies	MoSCoW
Workforce task allocation optimization	Medium	M	Workforce roster & shift data	Should Have
Shift dashboards with throughput & defect metrics	Medium	S	PLS-2 + production feed	Could Have

- **Phase 3 (12–18 months):** Optimization, energy management, and scale

Feature / User Story	Priority	Effort (S/M/L)	Dependencies	MoSCoW
Digital SOPs & AI-guided troubleshooting for new staff	Medium	M	Knowledge base, chatbot integration	Should Have
Factory digital twin for rapid reconfiguration to design changes	Medium	L	Simulation engine, live production data	Could Have
Energy monitoring & optimization aligned with off-peak pricing	Low	M	Machine energy sensors, local grid data	Could Have
Advanced analytics: cost per unit, defect trends, downtime patterns	Low	S	Data warehouse + BI dashboards	Could Have

4.7. Milestone List

Phase 1

- Sensor and ERP/MES integration completed
- Predictive maintenance model live
- Dynamic scheduling engine deployed
- Quality inspection MVP validated
- Pilot go-live and user training

Phase 2

- Supply chain forecasting and procurement automation live
- Vendor risk scoring implemented
- Workforce optimization dashboards released

Phase 3

- Energy optimization module live
- Advanced analytics and optimization completed
- Production-ready v1.0 release

4.8. Roles and Responsibility Matrix

Role	Key Responsibilities
Product Manager	Vision, scope, prioritization, stakeholder alignment
AI/ML Engineer	Model development, training, and optimization
Backend Engineer	Data ingestion, integrations, scheduling engine
Frontend Engineer	Dashboards and user interfaces
QA Engineer	Functional, performance, and reliability testing
UX Designer	User workflows, usability, and validation
Operations SME	Domain validation and pilot feedback
DevOps Engineer	Deployment, monitoring, and reliability

4.9. Tools and Tech Stack List

AI & Data

- Python, TensorFlow / PyTorch
- OpenCV for vision-based inspection

Backend

- Node.js / Python APIs
- Kafka or MQTT for sensor data ingestion

Frontend

- React-based web dashboards
- Mobile-responsive UI for supervisors

Infrastructure

- Cloud deployment (AWS / Azure)
- Docker, Kubernetes

Integration

- ERP/MES connectors (REST, OPC UA, MQTT)

4.10. Demo Plan

Audience: Plant leadership, operations, and technical stakeholders

Demo Flow

1. Live production dashboard overview
2. Simulated machine failure prediction
3. Auto-rescheduling in response to failure
4. Real-time defect detection example
5. Inventory risk alert and auto-PO trigger

Success Indicator
Stakeholders understand value within 15 minutes and can map it to their operations.

4.11. Final Delivery Checklist

- All in-scope features implemented and tested
- Performance and reliability benchmarks met
- Security and role-based access validated
- User training completed
- Pilot success metrics achieved
- Documentation and support handover completed
- Go/no-go sign-off from stakeholders

Component 5: Viability Check

5.1. Market Demand Assumption

Drone electronics manufacturing is scaling rapidly across defense, logistics, and industrial sectors. Manufacturers face increasing pressure to deliver higher volumes with limited skilled labour and tighter delivery commitments. Recurring downtime, quality rework, and component shortages directly impact revenue and customer trust. This creates sustained demand for AI-driven operational platforms that improve predictability without disrupting existing ERP or MES investments.

5.2. Willingness-to-Pay Assumption

Target customers are willing to pay for solutions that demonstrate clear operational ROI within a short timeframe. Operations leaders prioritize tools that reduce downtime, improve on-time delivery, and lower rework costs.

Assumption: If DroneOps AI delivers measurable improvements within 60–90 days, customers will approve recurring spend as part of operational expenditure rather than capital expenditure.

5.3. Revenue Model Selection

A **pilot-led, subscription-based model** is adopted:

- **Pilot Engagement:** Fixed-fee, ROI-tied (8–12 weeks)
- **Production Subscription:** Recurring annual subscription priced per plant or production line
- **Expansion Revenue:** Additional modules (quality, supply chain, energy optimization) priced as add-ons

This model aligns revenue with customer scale and realized value.

5.4. Pricing Assumption (Indicative)

- Pilot: ₹15–25 lakhs per plant
- Base subscription: ₹4–6 lakhs per production line per year
- Add-on modules: ₹1–2 lakhs per module per year

Pricing is positioned at less than 10–15% of the typical annual cost of downtime and rework for mid-sized plants.

5.5. Cost Estimate

Major Cost Drivers

- AI/ML development and model training
- Sensor and camera integration (where required)
- Cloud infrastructure and data storage (***localized***)
- Ongoing model monitoring and support

Estimated Cost Structure

Fixed Costs

- Core platform and AI model development
- Integration framework and scheduling engine
- Initial product management and UX

Variable Costs (per customer)

- Cloud compute and data storage
- Model retraining and monitoring
- Customer support and minor customization

5.6. Resource Availability Check

Available

- AI/ML engineers with predictive analytics and vision expertise
- Full-stack engineers for dashboards and integrations
- Manufacturing domain experts for validation

Dependencies

- Access to machine sensor data and historical production records
- Customer willingness to support pilot deployments

No critical resource gaps identified for MVP and early rollout.

5.7. Operational Feasibility Check

The solution is operationally feasible because:

- It layers on top of existing ERP, MES, and IoT systems
- Deployment can begin with a limited pilot scope
- Human-in-the-loop controls reduce operational risk

Operational complexity increases gradually as features expand, allowing phased adoption.

5.8. Competitive Viability Assessment

Competitive Landscape

- Traditional MES and ERP tools offer visibility but limited predictive action
- Point AI tools focus on single problems (maintenance or quality)
- Large industrial AI platforms are costly and slow to deploy

Competitive Advantage

- Integrated, agentic approach across operations, quality, and supply chain
- Faster time to value through pilot-led deployment
- Explainable AI with human override, tailored for manufacturing trust

Assumption: Customers prefer targeted, high-impact operational AI over large, generic platforms.
Data protection issues when dealing with generic platform corporations, and data-sharing limitations. More specific customer related solutions can be provided as add-ons with our platform including eventual scaling if needed.

Component 6: Use of AI

6.1. Identified AI Use Cases

1. Predictive maintenance for critical manufacturing equipment
2. Dynamic production scheduling and auto-rescheduling
3. Real-time quality inspection using computer vision
4. Supply chain demand forecasting and vendor risk prediction
5. Workforce task allocation and shift optimization

Among these, **dynamic production scheduling combined with predictive maintenance** is the **primary agentic AI use case**, as it requires continuous sensing, decision-making, and action.

6.2. GenAI / Agentic AI Use Case

Use Case: Agentic Operations Orchestrator

An agentic AI continuously monitors machine health, production progress, order priority, and inventory status, predicts disruptions, and autonomously adjusts production schedules while keeping humans in control.

6.3. Agent Responsibility

The Operations Orchestrator Agent is responsible for:

- Monitoring real-time machine, quality, and order signals
- Predicting imminent failures or bottlenecks
- Evaluating production priorities and constraints
- Generating optimized rescheduling recommendations
- Executing approved schedule changes automatically
- Explaining decisions with confidence scores and rationale

The agent does not replace human decision-making for critical shutdowns or compliance actions.

6.4. Agent Trigger

The agent is triggered when:

- Sensor data crosses failure risk thresholds
- Quality defect rates exceed defined limits
- Urgent or high-priority orders are added or modified
- Inventory levels fall below safety stock for critical components
- Manual override or “re-optimize” request is initiated by a user

Triggers can be real-time or batch-based depending on data source.

6.5. Agent Boundaries

The agent is explicitly constrained to:

- Operate only within predefined safety, quality, and compliance rules
- Modify schedules but not machine-level control parameters
- Execute actions only on approved machines, lines, or orders

- Require human approval for production halts, supplier switching, or workforce reassignment beyond thresholds
- Maintain audit logs for every decision and action

The agent cannot override regulatory, safety, or policy constraints.

6.6. Failure Handling

If the agent fails or confidence drops below acceptable levels:

- Fall back to rule-based scheduling or last known stable schedule
- Notify users with clear explanation of failure or uncertainty
- Suspend autonomous actions and switch to recommendation-only mode
- Log failure events for review and model improvement

System continuity is prioritized over aggressive automation.

6.7. AI Success Metrics

Prediction Accuracy

- ≥85% accuracy in failure and bottleneck prediction

Operational Impact

- ≥30% reduction in unplanned downtime
- ≥40% of production batches auto-rescheduled by AI

Quality Outcomes

- ≥25% reduction in rework and defect rates

Trust and Adoption

- ≥70% of recommendations accepted or executed
- ≥70% weekly active usage by target users

Explainability

- 100% of AI actions accompanied by rationale and confidence score

Component 7: Demo/Prototype

The below wireframes are created in FIGMA make

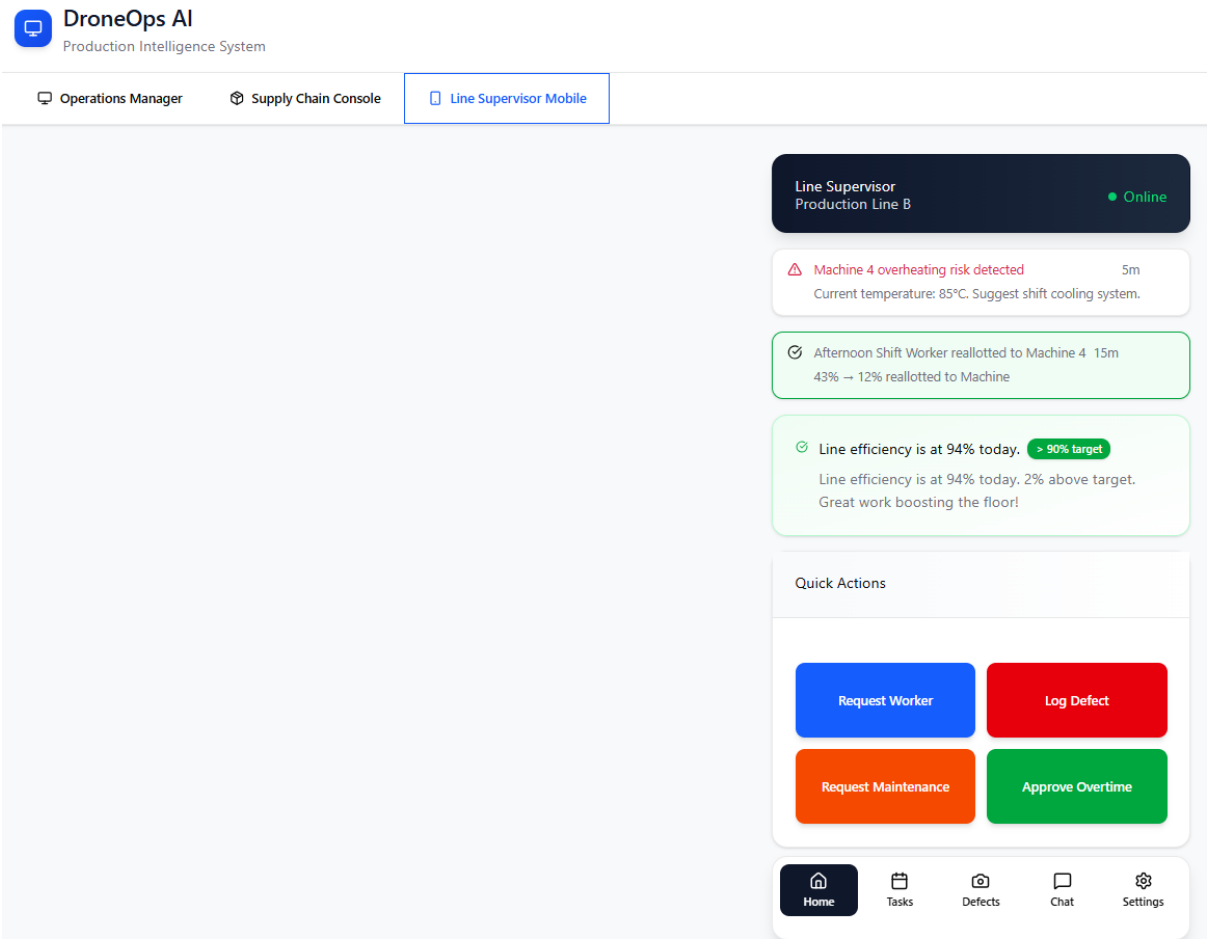
URL: https://sleek-sample-46864589.figma.site/

Password: Agenticace@321a

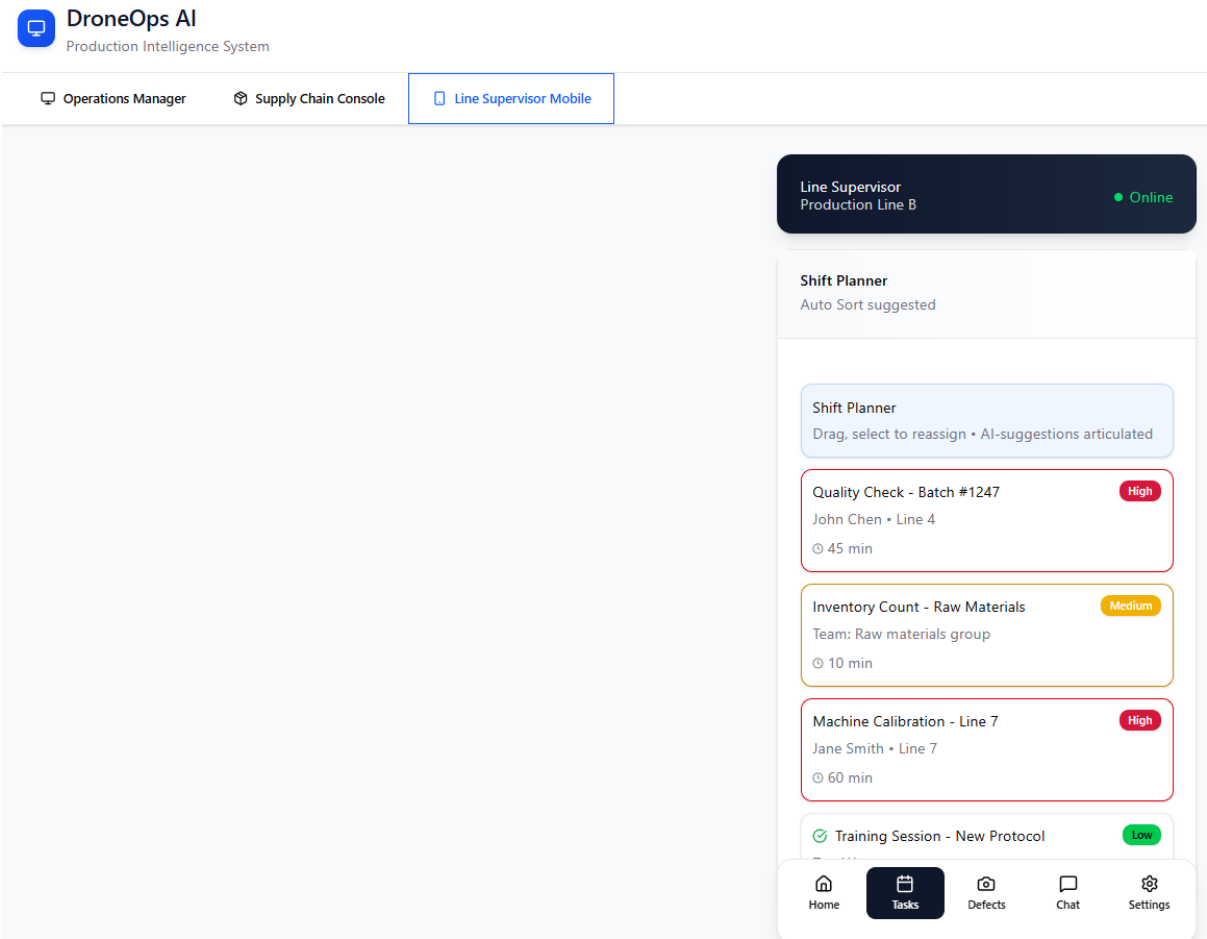
Wireframes:

1> Line Supervisor Mobile App Screenshots

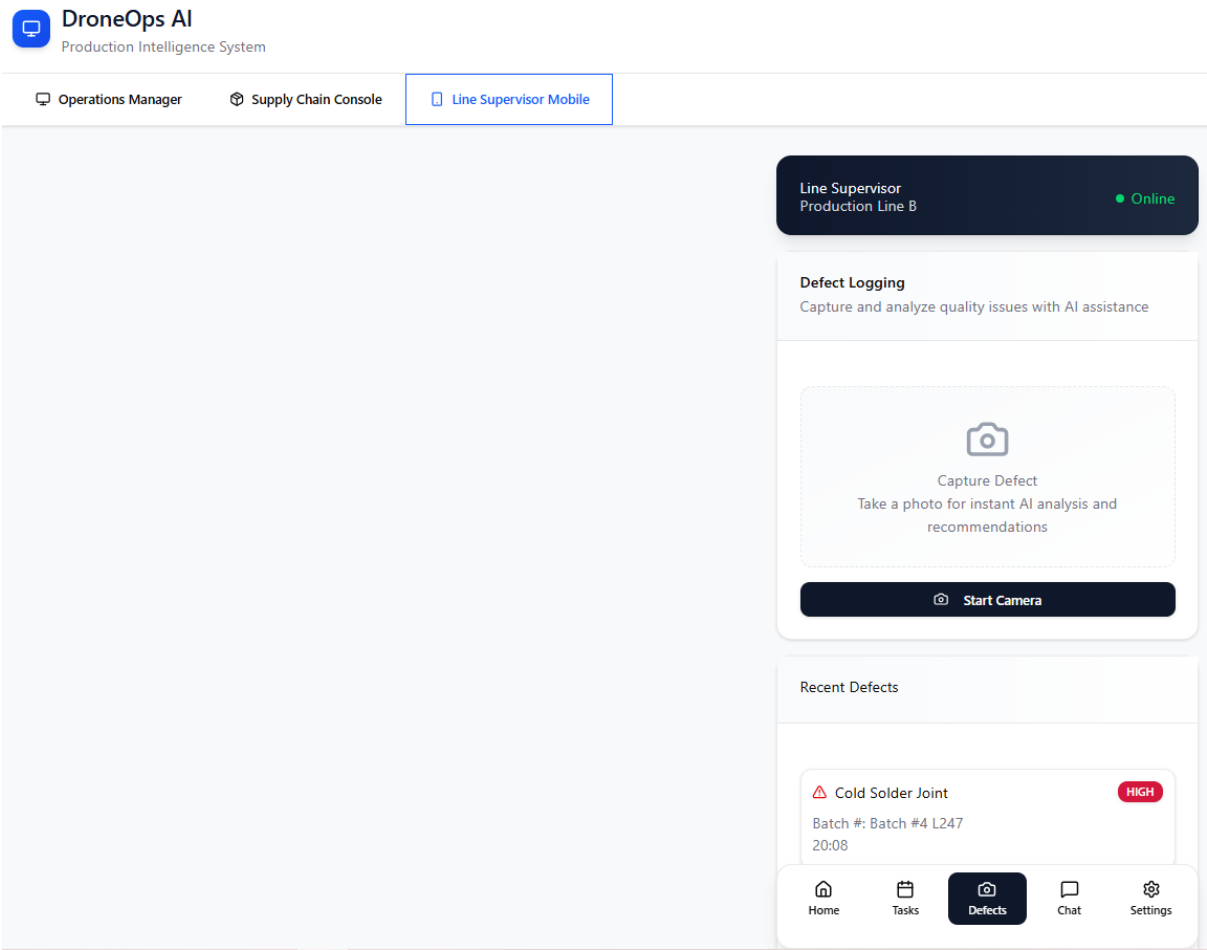
1.1> Home screen



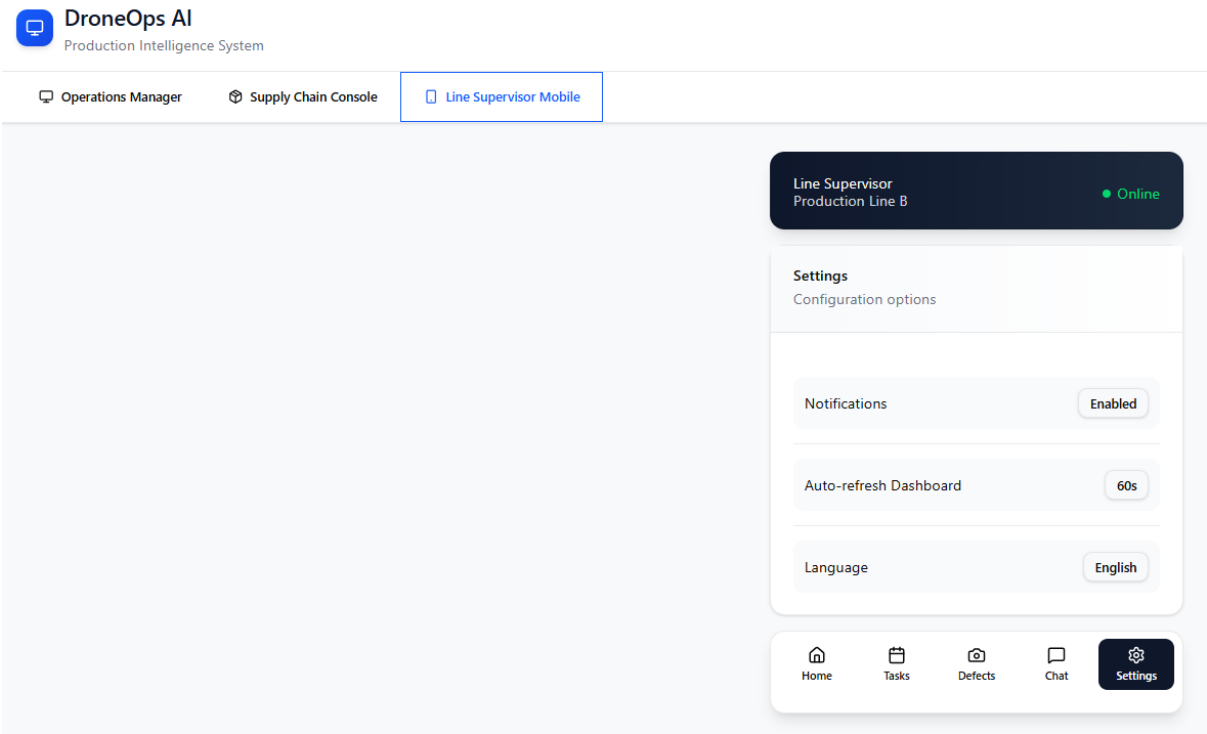
1.2> Task screen



1.3> Defects screen

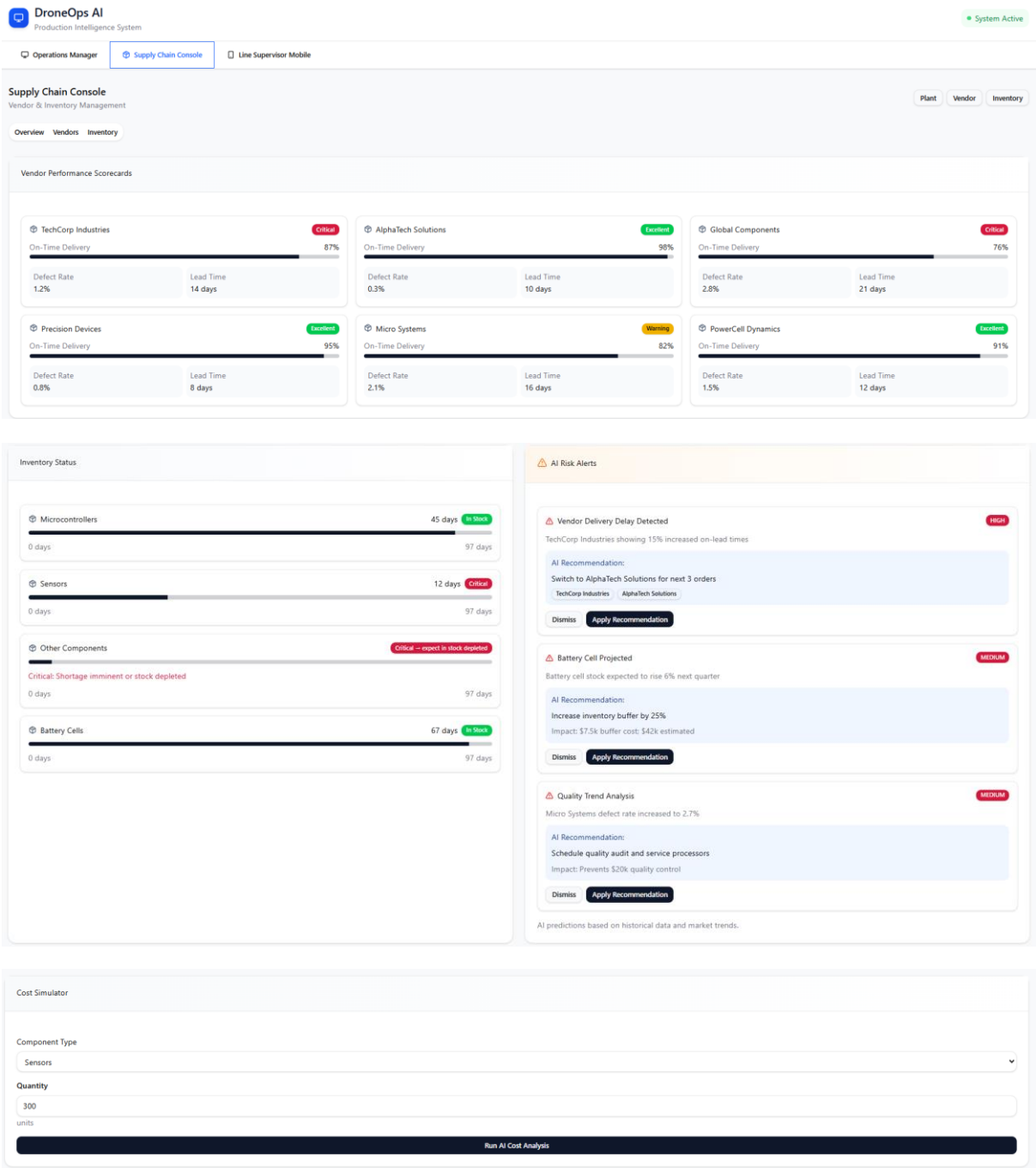


1.3> Settings screen

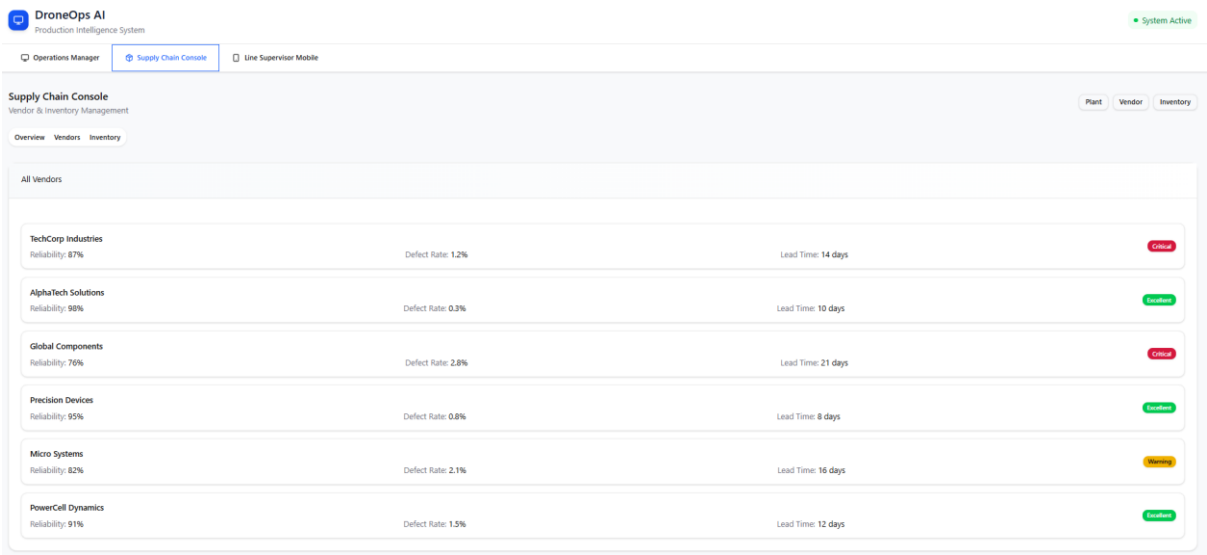


2> Supply Chain Console Screenshots

2.1> Overview screen



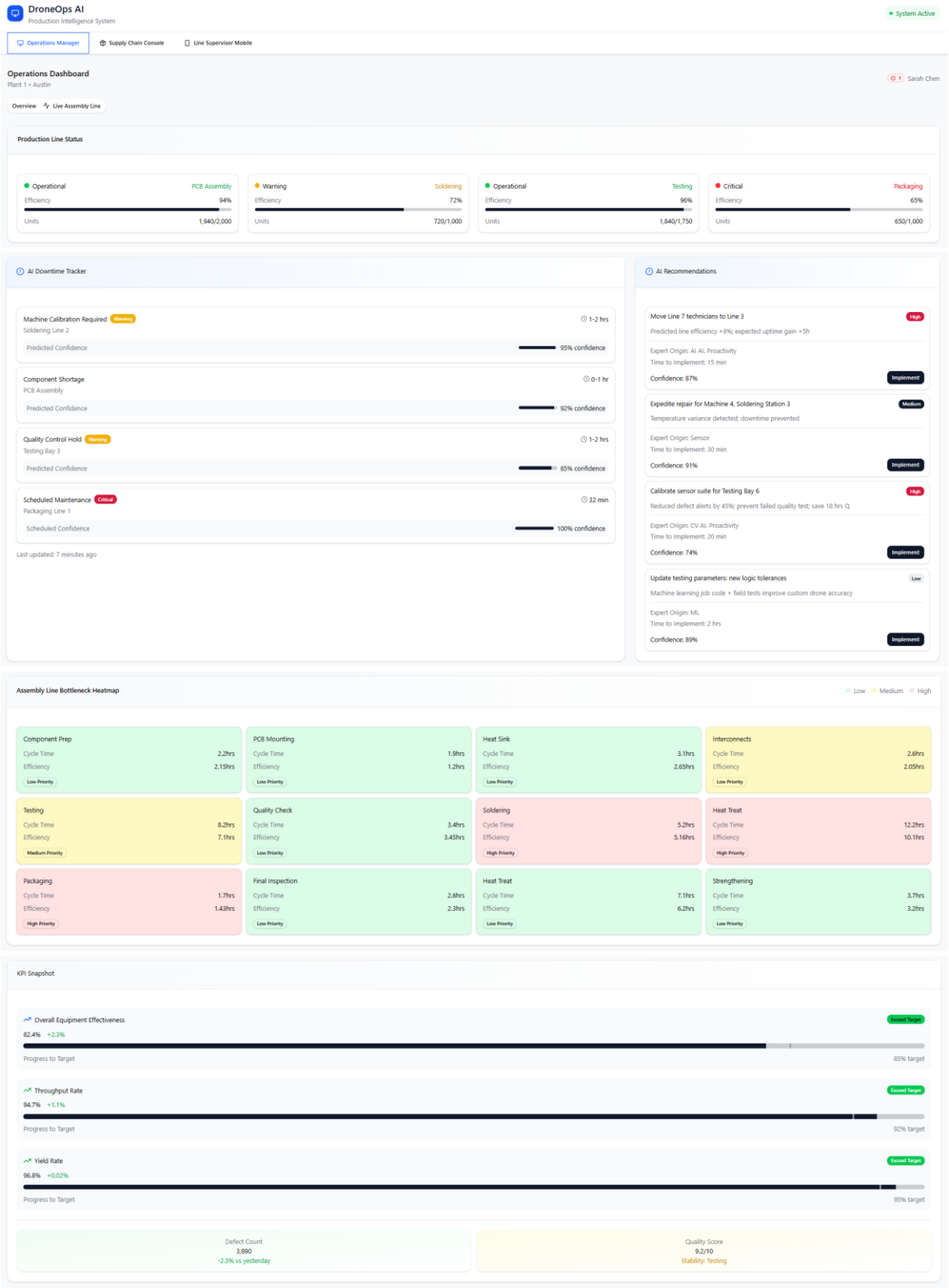
2.2> Vendors screen



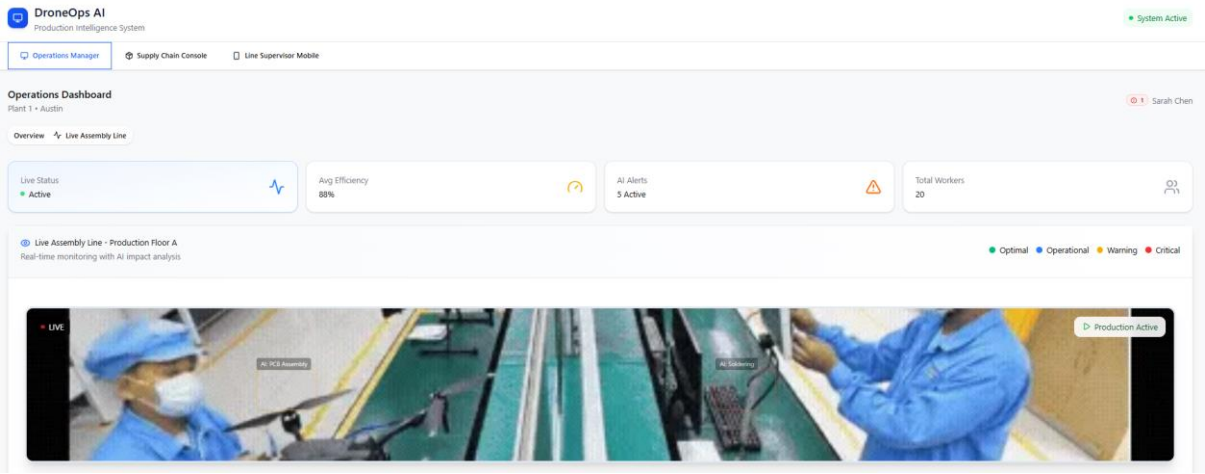
2.3> Inventory screen

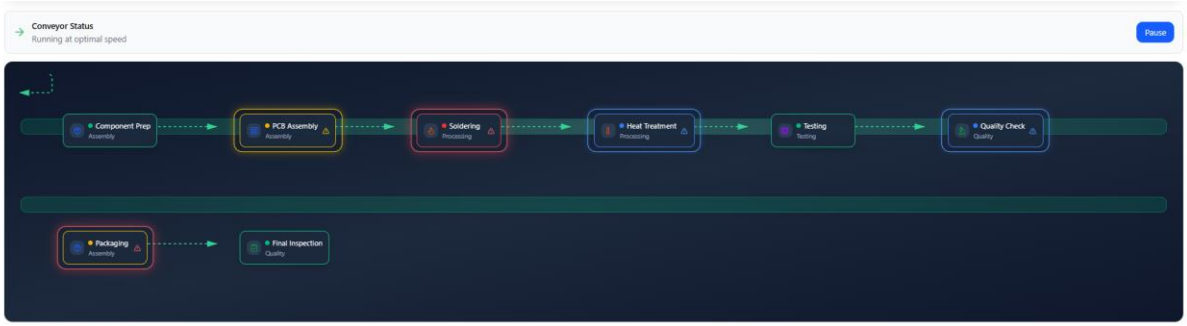
3> Operations Manager Dashboard

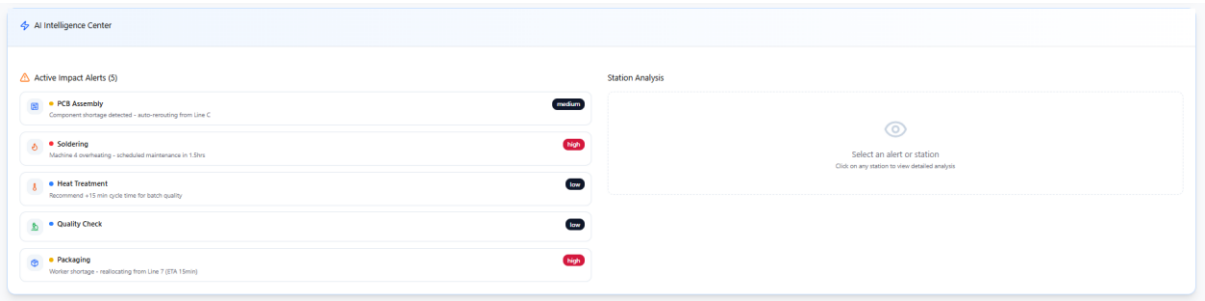
3.1> Overview screen



3.2> Live Assembly Line screen







Component 8: Success Metrics

8.1. User / Customer Metrics

Net Promoter Score (NPS)

- Measures user satisfaction and advocacy
- Target: ≥40 within 6 months of rollout

Daily Active Users / Monthly Active Users (DAU/MAU)

- Measures habitual usage by operations leaders
- Target: ≥0.5 for core personas

Acquisition Rate

- % of pilots converting to paid subscription
- Target: ≥60% pilot-to-rollout conversion

Activation Rate

- % of onboarded users completing first key workflow
- Example: Viewing alerts, approving a reschedule
- Target: ≥75% within first 14 days

8.2. Business & Revenue Metrics

Annual Recurring Revenue (ARR)

- Subscription revenue from active plants

Revenue per Plant / Line

- Measures monetization depth
- Target: Steady increase through module expansion

Customer Lifetime Value (LTV)

- Derived from average contract value and retention

Customer Acquisition Cost (CAC)

- Sales and onboarding cost per customer

LTV / CAC Ratio

- Target: ≥3 for sustainable growth

8.3. Product Usage & Engagement Metrics

Weekly Active Users (WAU)

- Measures ongoing relevance to daily operations
- Target: ≥70% of target users

Feature Adoption Rate

- % of users actively using predictive maintenance, scheduling, or quality modules

Alert Action Rate

- % of AI alerts that result in user action
- Target: ≥70%

Time to First Value

- Time from onboarding to first measurable improvement
- Target: ≤30 days

8.4. AI Performance Metrics

Prediction Accuracy

- Accuracy of failure, defect, and risk predictions
- Target: ≥85%

False Positive / False Negative Rate

- Ensures trust and operational relevance
- Target: <10% combined

Confidence Calibration

- Alignment between predicted confidence and actual outcomes

Model Drift Detection

- Frequency and magnitude of performance degradation

8.5. Growth Metrics

Customer Expansion Rate

- Increase in number of lines, modules, or plants per customer

Retention Rate

- Annual customer retention
- Target: ≥90%

Churn Rate

- Target: <10% annually

Pipeline Growth

- Qualified pilots entering funnel

8.6. Quality, Trust & Reliability Metrics

System Uptime

- Target: ≥99.9%

Alert Latency

- Time from event detection to alert
- Target: ≤60 seconds

Recommendation Acceptance Rate

- % of AI recommendations approved or executed
- Target: ≥70%

Override Rate

- Monitored to detect trust or usability issues

8.7. AI / Data / Automation Metrics

Autonomous Action Rate

- % of actions executed automatically within guardrails

Human-in-the-Loop Rate

- % of decisions requiring human approval

Data Freshness

- Time lag between sensor data generation and AI processing

Data Coverage

- % of critical machines and processes connected

Automation Success Rate

- % of automated actions executed without rollback

8.8. Overall Product Success Metrics

Quantitative

- ≥30% reduction in unplanned downtime
- ≥25% reduction in defects and rework
- ≥20% improvement in on-time delivery
- ≥15% reduction in energy consumption per batch

Qualitative

- User feedback indicates reduced firefighting and stress
- Leadership perceives improved predictability and control
- Increased trust in AI recommendations over time

Component 9: Presentation

Presentation is attached

DroneOPS AI Presentation.pdf

Ethical AI

This product applies agentic AI to drone electronics manufacturing, where decisions directly impact safety, quality, compliance, and workforce outcomes. Ethical AI is therefore a core product requirement, not an afterthought.

Human-in-the-loop control:
AI agents provide recommendations, predictions, and automated actions only within clearly defined guardrails. Critical decisions affecting production shutdowns, workforce reallocation, supplier switching, or compliance sign-offs require explicit human approval. Users can always override AI recommendations, and the system records rationale for traceability.

Transparency and explainability:
All AI-driven insights such as downtime predictions, quality flags, or supplier risk alerts are accompanied by clear explanations, confidence scores, and underlying signals (machine data, historical trends, vendor performance). This ensures operational teams trust the system and can validate decisions in regulated environments.

Data privacy and security:
The platform follows privacy-by-design principles. Only operational and machine data required for optimization is collected. Customer IP, design files, and sensitive supplier data are encrypted at rest and in transit, with strict role-based access control. The system complies with applicable Indian data protection regulations and enterprise security standards.

Bias and fairness:
AI models used for workforce scheduling and performance insights are continuously monitored to avoid bias against individuals, shifts, or teams. Recommendations focus on process optimization, not punitive evaluation of workers. Model outputs are regularly audited for fairness and unintended patterns.

Safety and compliance alignment:
AI agents are constrained by predefined safety, quality, and regulatory rules (e.g., DGCA, ISO, IPC standards). The system will not recommend actions that violate compliance thresholds, even if they appear operationally efficient.

Continuous monitoring and governance:
Model performance, drift, and outcomes are reviewed regularly. Ethical guidelines, audit logs, and escalation paths are built into the platform to ensure responsible, reliable, and compliant AI operations at scale.

Power and Resource Minimisation:
Switching to most viable energy source (centralized grid, DG, Solar) based on the work order variability, time of day, costing rates, and energy availability is done. Efforts are made to maintain minimum energy mode in machining stations to reduce energy losses. Automation is brought in for off-time orders where sufficient workforce isn't available (over the traditional over-time) so as to allow for work-life balance for them as well as maintaining output quality efficiency. Efforts are made to ensure a balance between human resources, automation, eccentric order completion to maintain well-wish between stakeholders & generating stably increasing long-term profits.

Go-To-Market (GTM) Strategy

Who We Target

Primary Customers:
Mid–large manufacturing plants (Automotive, Electronics, Aerospace, and Industrial).

Key Buyers:
Plant Head | VP Operations | CIO | Quality Head | Supply Chain Head.

Beachhead:
Digitally mature plants already using ERP, MES, and IoT.

What We Sell (Value Proposition)

"From alerts to action. AI that runs the plant with you."

- Reduces downtime, defects, and supply risk
- Agentic AI with human-in-the-loop control
- Works on top of existing systems
- ROI visible in every workflow
- Help to retain your top workers by allowing them work-life balance while still maintaining profits.

How We Enter (Use-Case Led)

Start small. Prove fast. Expand wide.

1. Predictive Maintenance
2. AI Vision Quality Inspection
3. Supply Chain Risk & PO Automation

Each use case is independent and expandable.

How We Sell (Motion & Pricing)

Sales Motion: Enterprise, consultative, pilot-first
Deal Cycle: 3–6 months

Pricing Model: Land & Expand

- **Pilot (8–12 weeks):** Fixed fee, ROI-based success metrics
- **Plant Rollout:** Subscription per plant / line
- **Enterprise:** Multi-plant, volume pricing

Why We Win (Differentiation)

- AI that recommends *and executes*
- Confidence scores and explainability
- Always-on human override
- Built for operational trust, not demos
- Closed loop AI models per plant / per Company that allows customization, secrecy, & high efficiency.

How We Scale (Channels & Expansion)

Partners: ERP, MES, IoT vendors, System Integrators
Expansion Path: Ops → Quality → Supply Chain → Leadership Analytics

GTM Success Metrics

- Pilot-to-rollout conversion
- Time to first ROI
- Expansion revenue per customer
- Downtime and defect reduction

Future Scope

Handling Multi-variety Orders (upto 3) Simultaneously

- 3 different types of drones and their support systems can be manufactured and integrated, even if the orders arrive at similar times. Selective automation and work variability only for the most efficient operators. Helping staff can be obtained on a freelance/contract basis with automated intimation (***planning agent***).
- Machine loading & resting times will be adjusted for almost continuous production without breakdown (weekly repair & maintenance day). Low speed & power modes for generic drone production orders (***maintenance agent***).
- Waste material from high end drones, will be utilized for the low end/generic usage to further reduce material wastage (***allocation agent***).

3 Variable Speed Production Rates

- According to delivery flexibility, machine loads & availability, & contract operator presence, 3 zones of production (quick-medium-minimal) will be set. Within ranges, work-flow & throughput will be adjusted (taking into consideration permanent operator efficiency & machining needs) (***variability agent***).
- In orders with the same customer and a few drones out of the order needing custom configurations, variable logging in production circuits will be achieved with minimal bottlenecking (***complementary add-on agent***).
- Energy load & availability-based speed switching to save on power costs with grid switching (***power-damping agent***).

Scale-up Agent (for regular & genuine clients)

- An approach to give end-to-end solutions to a client with multiple factories (in a series), where manufacturing & integration of almost all aspects is in-house (***integration + timeline + process flow agents***).
- For clients who plan on replicating their production models at different locations for mass production applications (***closed-loop agentic framework + motion agents***).
- For fixed contract operators who need to do manual work & tasks at multiple points (2-3) on the same production line, intimation & resource availability agents that aid in co-ordination (***intimation + multiplier agents***).

References

Sources:

[1] Drones can boost India's manufacturing potential to $23 billion by 2030: Report - The Economic Times

https://economictimes.indiatimes.com/tech/technology/drones-can-boost-indias-manufacturing-potential-to-23-billion-by-2030-report/articleshow/122299767.cms?from=mdr

[2] India Predictive Maintenance Market Size, Forecast 2033

https://www.imarcgroup.com/india-predictive-maintenance-market

[3] Drone Market

https://www.grandviewresearch.com/horizon/outlook/drone-market/india?utm_source=chatgpt.com

Glossary

Acronym	Full Form
AI	Artificial Intelligence
GenAI	Generative Artificial Intelligence
JTBD	Jobs To Be Done
PRD	Product Requirements Document
MVP	Minimum Viable Product
GTM	Go To Market
ERP	Enterprise Resource Planning
MES	Manufacturing Execution System
IoT	Internet of Things
PLC	Programmable Logic Controller
SOP	Standard Operating Procedure
API	Application Programming Interface
NPS	Net Promoter Score
DAU	Daily Active Users
MAU	Monthly Active Users
WAU	Weekly Active Users
ARR	Annual Recurring Revenue
CAC	Customer Acquisition Cost
LTV	Lifetime Value
ROI	Return on Investment
KPI	Key Performance Indicator
BI	Business Intelligence
CV	Computer Vision
OPC UA	Open Platform Communications – Unified Architecture
MQTT	Message Queuing Telemetry Transport
DGCA	Directorate General of Civil Aviation (India)